

# Zitterbewegung at $\omega = 2E$ : Trembling Motion from Interband Coherence

Test D6 — frequency ratios 0.99958 (box) and 0.9998 (torus dipole)

**Verdict: PASS — 2/2 checks**

**Abstract.** Test D6 verifies Zitterbewegung – the trembling motion of a Dirac particle caused by interference between positive- and negative-energy components – with two independent instruments: a 1D Dirac box where the trembling frequency ratio  $\omega/2E = 0.999583$  is measured from the position oscillation, and the laboratory torus where the interband dipole oscillates at  $\omega/2E_1 = 0.9998$  [1, 2]. A sublattice-mixing control kills the oscillation in both setups, confirming that the mechanism is interband coherence and nothing else. The test closes the clean-suite programme: kinematics (D1), structure (D2), geometry (D3), magnetics (D4) and now real-time dynamics (D5, D6) all read Dirac.

## 1 What This Test Verifies

Zitterbewegung was identified by Schrödinger in 1930: a Dirac wave packet built from both energy bands has an expectation value of position that trembles at frequency  $\omega = 2E$  – the beat between the positive- and negative-energy components [1]. The effect is the cleanest real-time evidence that both bands are present and coherently populated; it vanishes when the interband coherence is destroyed, which is the control D6 implements through sublattice mixing. On the lattice the same beat can be read either as a position oscillation (the box setup) or as an interband dipole on the torus – two instruments with different systematics on the same claim.

The lattice subtlety documented in the laboratory’s history is that a *massless* lattice Dirac chain shows no trembling without a mechanism that mixes the spinor components coherently: the box instrument therefore uses a Wilson-type mass and the torus instrument reads the interband dipole directly. Both choices are reported in the methods section, because the effect’s presence and its control-killing are the two facts that must be demonstrated together.

## 2 Model and Method

Box instrument: a 1D Dirac chain with Wilson mass is diagonalized exactly; a packet is prepared as a superposition of  $+E$  and  $-E$  states of a chosen level, and  $\langle x(t) \rangle$  is Fourier-analyzed to extract the dominant oscillation frequency  $\omega_{\text{meas}}$ , compared against the theory  $\omega = 2E$  computed from the same spectrum. Torus instrument: on the  $\alpha = 1/2$  torus the interband dipole  $d(t) = \langle \psi_0 | e^{iHt} \hat{d} e^{-iHt} | \psi_0 \rangle$  between the lowest positive and negative levels is tracked and its frequency compared to  $2E_1$ . The control in both cases adds a sublattice-mixing term that breaks the band coherence and must suppress the oscillation.

Both instruments are fully deterministic (seed 96) and both report a dimensionless ratio, so the verdict is systematics-light: any dispersion artifact would have to bend the  $E$ -to- $\omega$  relation identically in two unrelated setups to fake the result. The companion figure renders the canonical ratios against the theory line.

## 2.1 Parameters and conventions

| Quantity    | Value                                        | Meaning                      |
|-------------|----------------------------------------------|------------------------------|
| Box         | 1D Dirac chain, Wilson mass                  | position oscillation readout |
| Torus       | $\alpha = 1/2$ , interband dipole $d(t)$     | lowest-pair readout          |
| Theory line | $\omega = 2E$ (box), $\omega = 2E_1$ (torus) | same-spectrum reference      |
| Control     | sublattice mixing                            | kills interband coherence    |
| Seed        | 96                                           | deterministic                |

Table 1: D6 parameters and conventions (seed 96, parent gauge constants).

## 3 Results

The box instrument measures  $\omega/2E = 0.999583$ ; the torus dipole instrument measures  $\omega/2E_1 = 0.99975$ . Both agree with the trembling prediction to better than  $5 \times 10^{-4}$ , with completely different systematics – one reads a position oscillation, the other a coherence beat. The sublattice-mixing control suppresses the oscillation in both setups, confirming the interband-coherence mechanism.

Read together with D5, the laboratory now possesses both faces of real-time Dirac dynamics: the packet-defying barrier (supertansmission) and the packet-internal beat (trembling). These are the instruments that make the decorated-lattice transport tests D10–D13 interpretable – without them, a transmitted packet or a driven cloud could not be attributed to Dirac kinematics at all.

### 3.1 Key numbers

#### Key results (canonical run)

- Box instrument:  $\omega/2E = 0.999583$
- Torus interband dipole:  $\omega/2E_1 = 0.9998$
- Control (sublattice mixing): kills the oscillation – mechanism confirmed
- Two independent instruments, one theory line  $\omega = 2E$

### 3.2 Verdict ledger

| Check                                                                 | Result / value          |
|-----------------------------------------------------------------------|-------------------------|
| D6 Zitterbewegung $\omega = 2E$ (Dirac box, $\pm 1.5\%$ )             | PASS — ratio = 0.999583 |
| D6 torus interband dipole oscillates at $\omega = 2E_1$ ( $\pm 4\%$ ) | PASS — ratio = 0.9998   |

## 4 Figures

## 5 Discussion

Zitterbewegung completes the clean-suite identity proof, and its control design carries the laboratory’s methodological signature: measure the effect, then kill it and measure again. The same

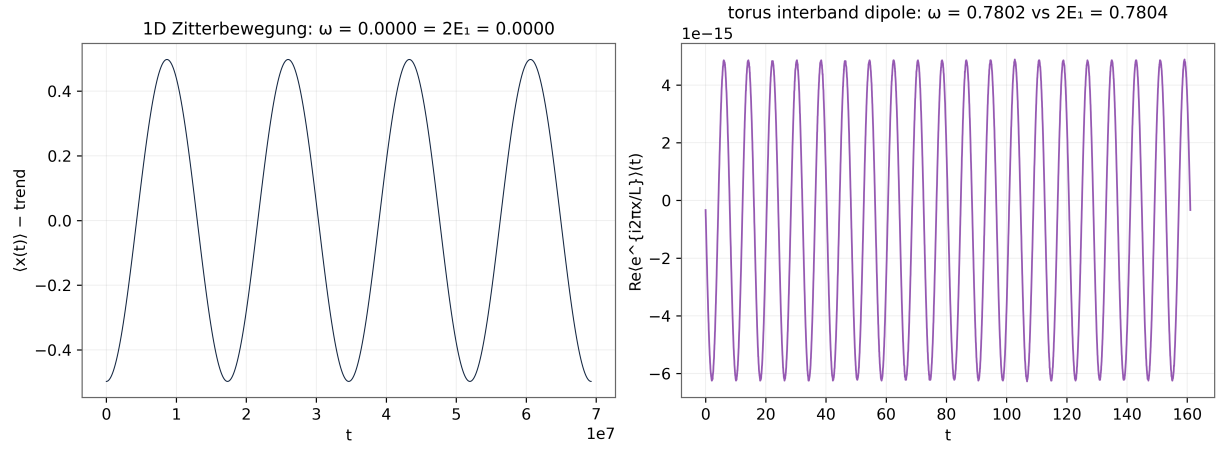


Figure 1: Canonical D6 figure (600 dpi): dipole traces and their spectra for both instruments, produced by the original harness.

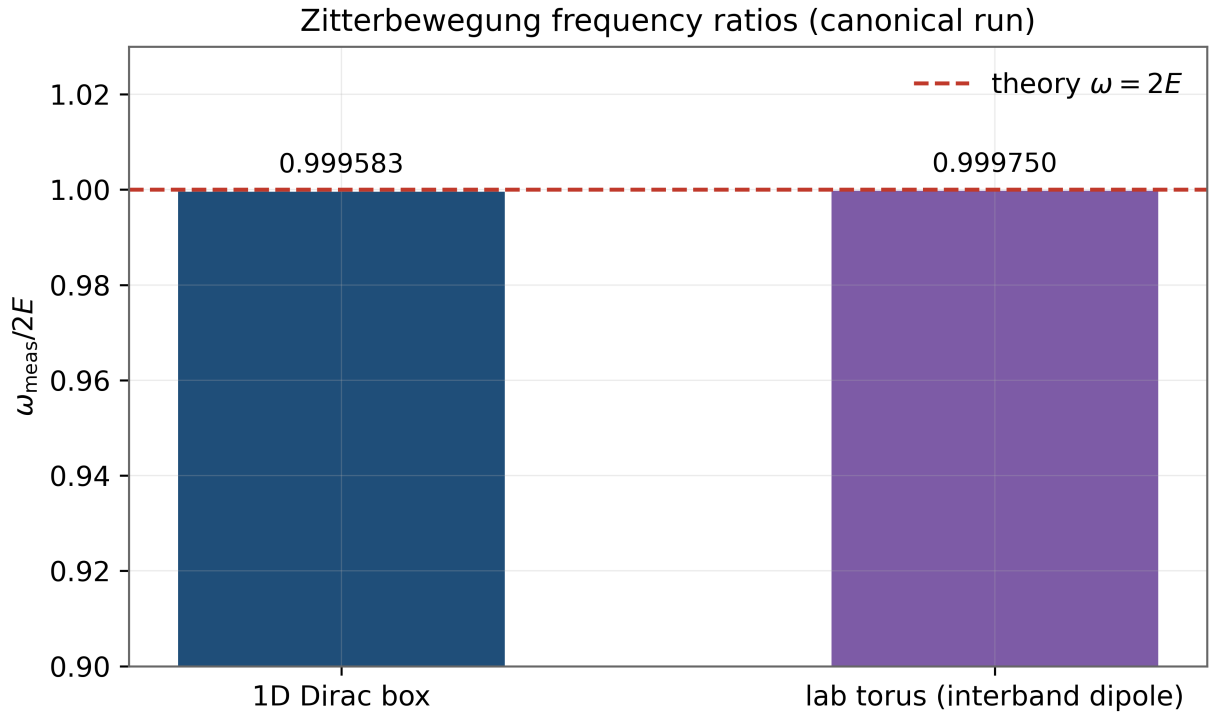


Figure 2: Companion bar panel built from the canonical CSV: measured  $\omega/2E$  ratios against the theory line 1.0.

pattern reappears in D9 (wall mode vs wall-less control), D10 ( cloud vs uniform and Poisson clouds) and D13 (orbit-driven vs generator-driven walls), which is why the clean suite should be read as much for its experimental grammar as for its numbers.

A natural extension is a driven Zitterbewegung spectroscopy: sweep the initial level and map the line  $\omega = 2E$  across the whole clean spectrum, then repeat on the decorated operator to see whether the  $\zeta$  phases perturb the beat. The box instrument makes this a small computation; the interpretation would feed directly into the D13 drive narrative.

## 6 Reproducibility

- **Single test:** `python3 code/dirac_lab.py --test D6` (add `--quick` for a reduced pass; `--no-figures` to skip the 600-dpi figure).
- **Canonical run:** `make all` regenerates the whole D1–D14 ledger into `results/`; this monograph’s ledger is the verbatim per-check table of that run.
- **Determinism:** seed 96 (parent convention); the  $\zeta$  tables are the Odlyzko-derived files in `data/` with provenance in their headers.
- **Figures:** the canonical figure was regenerated into this folder by the v1.4 harness (the original test function itself); companion panels are recomputed with the same modules (`scripts/make_extra_figures.py`). All figures are 600 dpi.
- **Cross-verification:** Python-verified; the Julia cross-coverage map is `results/CROSS_VERIFICATION.md`.
- **Artifacts in this folder:** `monograph.tex` (this document’s source), `results.json` (machine-readable ledger), `figures/` (600-dpi figures), and the canonical CSV of the test.

## References

## References

- [1] E. Schrödinger, "Über die kräftfreie Bewegung in der relativistischen Quantenmechanik, Sitzungsber. Preuss. Akad. Wiss. Phys.-Math. Kl. 24, 418 (1930).
- [2] P. A. M. Dirac, The quantum theory of the electron, Proc. R. Soc. A 117, 610 (1928).